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RESEARCH ARTICLE

Aggregating Contextual Information for Multi-Criteria Online Music Recommendations

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ABSTRACT This paper introduces CAMCMusic, a novel context-aware multi-criteria music recommendation system designed to address these limitations without relying on user-specific attributes, music features, or explicit user ratings. CAMCMusic integrates contextual information into a multi-criteria decision-making framework to align music genre recommendations with the user's contextual situation for generating Top-N music recommendations. CAMCMusic begins by assessing the relevance of contextual information based on the relationship between three key elements: the user, the music genre, and the user's context. Subsequently, we employ an aggregation technique to reveal the connection between context and overall ratings. On the MusiClef dataset, CAMCMusic achieves a precision@5 of 0.61, a recall@5 of 0.77, and F1@5 of 0.68, surpassing the next best performer, SVD++, by 52.5%, 11.6%, and 36%, respectively. On the CAL500 dataset, CAMCMusic outperforms the FM algorithm by 11.9% in precision, 16.4% in recall, and 20% in F1-Score. CAMCMusic significantly enhances recommendation outcomes by effectively capturing and utilizing the relationships between user contexts and music genres, thus providing a robust solution to the cold start problem while maintaining high recommendation quality and user satisfaction. The findings underscore the potential of combining context-awareness with multi-criteria decision-making (MCDM) to advance the state of the art in music recommendation systems.

INDEX TERMS Recommender systems, music, user-centric computing, multi-criteria decision, context-aware recommendation system, playlist continuation.

I. INTRODUCTION

Music is a universal human activity, present across all known societies [3]. It serves a diverse array of functions, both personal and social, and has been extensively studied across disciplines such as psychology, neuroscience, sociology, and computer science [1]. Understanding why people listen to music and how they develop their preferences is a shared interest among these fields. Research highlights several core functions of music listening, including self-awareness, mood regulation, and social relatedness [3]. The enjoyment of music is thus both a personal and social activity, with social motivations such as signaling group membership, fostering connections, and gaining social insights [4], [5]. These functions are fulfilled not only through shared experiences but also through solitary listening, a concept known as "social surrogacy" [6]. In the digital age, music streaming platforms have revolutionized how individuals access and experience music, offering on-demand access to millions of tracks [8]. These platforms extensively utilize recommender systems (RSs) to help users navigate vast music collections and discover tracks that align with their preferences [10], [11]. RSs mitigate the challenge of information overload by recommending items based on user data, typically involving user-item interactions and associated features [9]. However, recommending music poses unique challenges compared to other domains. The subjective nature of musical taste, the multifaceted attributes of music, and the influence of contextual factors such as time, location, and emotional state complicate the recommendation process [13], [14]. Users tend to listen to different types of music depending on their activities and often curate playlists tailored for specific contexts [15], [16]. Additionally, defining similarity between songs is challenging due to the high subjectivity involved

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in human evaluation [7]. Musical genres, while providing some guidance, have ambiguous boundaries and can overlap, making it difficult to establish objective measures of similarity [10], [17]. Traditional content-based and collaborative filtering approaches have limitations in capturing the deeper semantic information embedded within music, often overlooking connections between music and other entities like artists, albums, and playlists [21], [23]. Collaborative filtering methods require extensive user behavior data, posing challenges in scenarios involving new users or cold-start problems [24], [25]. Moreover, the data utilized for recommendations has become increasingly diverse, encompassing user ratings, music tags, and multi-modal data such as text, images, and audio [20], [21]. Effectively leveraging this diverse data remains a significant challenge. A common issue across most recommender systems, including music recommender systems (MRSs), is popularity bias [27], [28]. Popular artists receive more attention and are frequently recommended, while lesser-known artists struggle to gain exposure [29]. This bias can lead to dissatisfaction among users interested in discovering new music and contributes to the "rich get richer" phenomenon [30]. Addressing fairness in recommendation systems is a growing area of research, aiming to balance the interests of different stakeholders, including listeners and artists [31], [32]. In recent years, context-aware recommender systems (CARSs) have been developed to incorporate user contextual information, such as location, time, and activities, to generate more relevant recommendations [33], [34]. While CARSs expand the application scope of RSs, they often rely on predefined explicit contexts, which can raise privacy concerns, be resource-intensive to collect, and may not capture the dynamic nature of context [35]. To address these challenges, this study proposes a novel MRS called CAMCMusic, which integrates contextual information into the criteria definition through the application of multi-criteria decision-making (MCDM). Specifically, we focus on generating music playlist recommendations by ranking music genres that align with the user's current context [35]. Unlike traditional MRSs that rely on explicit user preference data, our approach leverages pre-collected data to bootstrap the system, addressing issues such as the cold start problem, diversity of recommended playlists, and scalability concerns. By aligning the user's current context with clusters of music genres sharing similar contextual dimensions, we aim to enhance the efficacy of music recommendations. Our Key contributions are:

- We measure the significance of contextual information in our proposed MCDM to identify factors influencing the listener's musical preferences.
- This study integrates context-awareness with MCDM within the domain of MRS.
- We conduct a comparative analysis of our proposed MRS approach against both content-based and existing context-aware algorithms, demonstrating the advantages of incorporating context and MCDM to enhance recommendations.

Our CAMCMusic study is organized as: Section II discusses the recommendation system, MCDA system, and context-aware recommendation systems in detail. We formulate the problem of context-aware MCDA MRS in Section III. Section IV elaborates on the different steps of the proposed CAMCMusic in great detail with their mathematical formation. The experiment step, comparative evaluation, and analysis are presented in Section V along with the findings and limitations of the work, and future direction in MRS research. Section VI summarizes our research.

II. LITERATURE REVIEW

This section reviews existing literature on music recommender systems (MRSs), focusing on traditional recommender systems, multi-criteria recommender systems, and context-aware recommender systems.

A. MUSIC RECOMMENDER SYSTEMS

MRSs have garnered significant attention from both academia and industry [7], [22]. With music streaming services like Spotify, Pandora, and Apple Music offering tens of millions of tracks, MRSs play a crucial role in filtering options and reducing choice overload [36]. However, traditional MRSs often fall short of user expectations due to their reliance on user-item interactions or content-based descriptors, without adequately considering factors influencing musical preferences [34].

To meet users' musical entertainment needs more effectively, it is essential to consider intrinsic factors (e.g., personality, emotional state) [38], [39], extrinsic factors (e.g., current activities) [40], [41], and contextual factors (e.g., weather, social surroundings, location) [33]. The composition and annotation of playlists can also provide insights into song compatibility and suitability for specific occasions [42], [43]. Therefore, a holistic approach is necessary for designing MRSs that are tailored to individual users.

The music domain presents unique challenges for RSs compared to other fields like movies or products. Music tracks are shorter, consumed in sequences, and often experienced passively or in varying contexts, requiring specific design considerations and the adoption of deep learning approaches [43], [44]. Advances in music information retrieval and audio signal processing have enhanced the role of content-based features, which deep learning techniques can effectively utilize [7]. Additionally, music's strong emotional impact necessitates emotion-aware MRSs that align recommendations with the user's mood. Deep learning techniques, such as recurrent neural networks, are effective in recommending ordered playlists rather than random songs [10].

Conventional MRSs are typically divided into content-based, collaborative filtering (CF), and hybrid approaches

[45], [46]. Content-based methods focus on the attributes of the music itself, while CF methods rely on user behavior data. However, CF methods face challenges with new users or cold-start problems [24], [25]. Hybrid approaches combine these methods to leverage their strengths. Recent studies have explored the use of knowledge graphs to enhance hybrid recommendation systems by enriching item descriptions with external data sources [46], and the use of deep learning models to incorporate textual information and improve track recommendations [47]. Clustering techniques have also been employed to enhance scalability and focus on clusters relevant to users' preferences [48], [49]. Addressing the challenge of users with unusual preferences (gray-sheep users) has been explored by identifying them based on listening habits to ensure more accurate recommendations [50].

B. MULTI-CRITERIA RECOMMENDER SYSTEMS

Multi-criteria recommender systems (MCRS) extend traditional RSs by describing items through multiple criteria and incorporating user feedback on these criteria [51], [52]. This allows for more refined recommendations, reflecting user preferences regarding different attributes of items. MCRS approaches are generally categorized into heuristic-based and model-based methods. Heuristic-based methods extend traditional similarity calculations to account for multi-criteria rating information by aggregating similarities derived from individual criteria [53], [54]. For example, [55] proposed a method that identifies neighbors of an active user by ranking preferences for item criteria and predicting overall ratings using an extended similarity-based approach. Similarly, [51] used genetic programming to aggregate user similarities and determine overall ratings.

Model-based methods, including the approach proposed in this paper, posit that overall item ratings are closely related to multi-criteria ratings [56], [57]. These methods construct predictive models to estimate a user's overall rating based on observed multi-criteria ratings. Model-based approaches are recognized for their flexibility and efficiency, as they can be applied across various recommendation techniques [56]. The "Criteria Chains" model [57] constructs a sequence of criteria, predicting user ratings for each criterion sequentially and estimating the overall rating through linear and conditional aggregations. Another approach uses a unified global and local tensor factorization to address multi-criteria recommendation, learning both a global predictive model and multiple local models [58]. Additionally, non-linear co-embedding models have been proposed for rating prediction by modeling user-item relationships as bipartite graphs and incorporating additional criteria and content features [59].

C. CONTEXT-AWARE RECOMMENDER SYSTEMS

Context-aware recommender systems (CARS) incorporate contextual information to enhance recommendation accuracy by considering users' situational environments [35], [37]. Contextual dimensions (e.g., season, trip type) and contextual conditions (e.g., winter, family trip) define the contextual situation [52]. Research in CARS can be broadly categorized into two approaches. The first integrates context into a single-criterion-based recommendation framework, assessing item relevance based on the user's current context [60], [61]. For instance, context-aware matrix factorization (CAMF) incorporates contextual rating deviations into traditional matrix factorization [62], with further developments considering contextual correlations using context similarity models [63]. CAMF models have been applied to generate point-of-interest (POI) recommendations by leveraging user preferences and real-time demand [64]. Venue-aware recommender systems for music have been proposed, inferring users' music preferences based on their venue and surroundings [60]. Model-based collaborative filtering methods have also been used to facilitate comprehensive POI recommendations for social users, identifying user similarities based on geo-tags from social media posts [65].

The second approach treats CARS as a multi-criteria decision-making (MCDM) problem, integrating MCDM methods with context-aware recommendations using criteria based on contextual information such as user activities or emotions [11], [52]. Studies have explored this integration in various domains, demonstrating its benefits in enhancing recommendation quality [53], [66].

D. INTEGRATION OF CONTEXT AND MULTI-CRITERIA DECISION MAKING

Despite the advancements in both MCRS and CARS, there are few studies that integrate these approaches within a single recommender system [11], [52]. [66] expanded the recommendation space by incorporating additional contextual information and multi-criteria ratings into the traditional user-item space, using a four-dimensional tensor and multilinear singular value decomposition to identify neighbors under relevant contextual information. Reference [67] integrated context into existing MCRS and applied them within educational learning, utilizing independent and dependent methods for multi-criteria rating prediction and linear and conditional aggregation methods for rating aggregation [35]. However, incorporating context in the rating aggregation process did not necessarily improve prediction accuracy.

Treating CARS as an MCDM problem offers a promising avenue for enhancing recommendation quality. By integrating MCDM methods with context-aware recommendations, using criteria based on contextual information, more accurate recommendations can be achieved [11], [52]. In the music domain, however, no existing work has integrated contextual information into the criteria definition, which has traditionally been limited to item attributes. This study aims to fill this gap by incorporating context within an MRS system through the application of MCDM, focusing on generating music playlist recommendations by ranking music genres that align with the user's context.

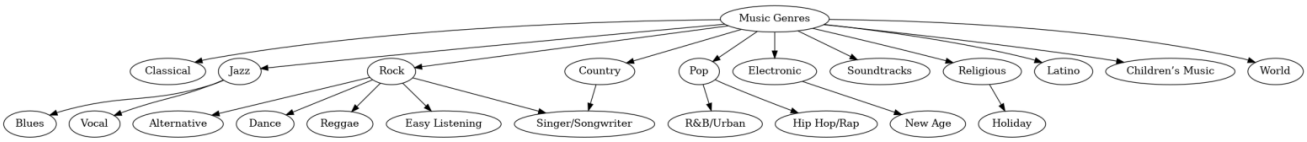

FIGURE 1. A hierarchical map categorizing a variety of music genres into their respective classifications.

III. PROBLEM FORMULATION

The core objective of this study is to design and develop a music context-aware recommender system that captures the nuanced relationship between users' preferences and their contextual situations. Specifically, the study aims to analyze whether individuals' music preferences align with their emotional states–such as choosing melancholic music when feeling down–or if they tend to select contrasting genres, such as upbeat tracks, to alter their mood. Additionally, the investigation explores inter-individual variability, acknowledging that preferences may differ significantly across users based on context-specific factors.

A. CONTEXTUAL FACTORS

The accurate representation of context is a pivotal step in developing an effective recommender system. Identifying domain-relevant contextual factors is crucial, as the inclusion of extraneous information can introduce noise, complicate the recommendation process, and degrade prediction accuracy [68].

To address this, we adopted a vector model to systematically integrate and represent contextual dimensions within the **CAMCMusic** framework. Each context is represented as a multidimensional vector, where individual dimensions correspond to specific contextual factors. These dimensions, defined statically, are associated with explicit attributes, such as temporal context (e.g., time of day or day of the week). By capturing contextual variations through a structured vector representation, the model accommodates the interplay between diverse contextual factors, such as the impact of both time and user activity on music preferences.

This vector-based structure supports scalability, allowing additional contextual dimensions to be seamlessly integrated as required for specific application scenarios. Formally, the vector model of context is defined as follows:

$$\mathcal{C} = (\mathcal{C}_T, \mathcal{C}_L, \mathcal{C}_P, \mathcal{C}_A, \mathcal{C}_E, \mathcal{C}_S) \tag{1}$$

where $\mathcal{C}_T$, $\mathcal{C}_L$, $\mathcal{C}_P$, $\mathcal{C}_A$, $\mathcal{C}_E$, and $\mathcal{C}_S$ denote the temporal, location, physical, activity, emotional, and social contextual dimensions, respectively.

1) IMPORTANCE OF CONTEXTUAL DIMENSIONS IN THE VECTOR MODEL

Each contextual dimension significantly contributes to the model's ability to deliver personalized and context-sensitive music recommendations:

a: TEMPORAL CONTEXT ($\mathcal{C}_T$)

Time-specific factors influence music preferences, such as selecting upbeat music for morning routines or relaxing tracks for evening wind-downs. Temporal adaptation allows the system to align recommendations with users' daily and weekly patterns.

b: LOCATION CONTEXT ($\mathcal{C}_L$)

User location impacts music selection, with preferences varying across settings, such as home, gym, or office. By considering location, the system enhances its adaptability to diverse environments.

c: PHYSICAL CONTEXT ($\mathcal{C}_P$)

Environmental factors like weather or noise levels influence user preferences. For instance, introspective music might be preferred on rainy days, while sunny weather might encourage cheerful selections.

d: ACTIVITY CONTEXT ($\mathcal{C}_A$)

Music preferences change with activities such as working, exercising, or relaxing. For example, instrumental music might suit studying, while energetic tracks align with workouts.

e: EMOTIONAL CONTEXT ($\mathcal{C}_E$)

Emotions play a critical role in shaping music choices. Whether reflecting a mood or attempting to alter it, emotional context enables the system to provide mood-congruent recommendations.

f: SOCIAL CONTEXT ($\mathcal{C}_S$)

Social dynamics influence preferences, with users opting for widely appreciated music in group settings and exploring niche genres in solitary contexts.

B. MUSICAL PREFERENCES

Music preferences can be modeled across multiple levels of abstraction, encompassing individual tracks, artists, albums, genres, and sub-genres. Each abstraction level captures unique insights into user tastes. For instance, preferences for specific songs provide granular data for precise recommendations, while genre-based preferences enable broader adaptability. Mood-related attributes, such as "Relaxing" or "Upbeat," offer further personalization by aligning suggestions with the user's emotional state and context [10].

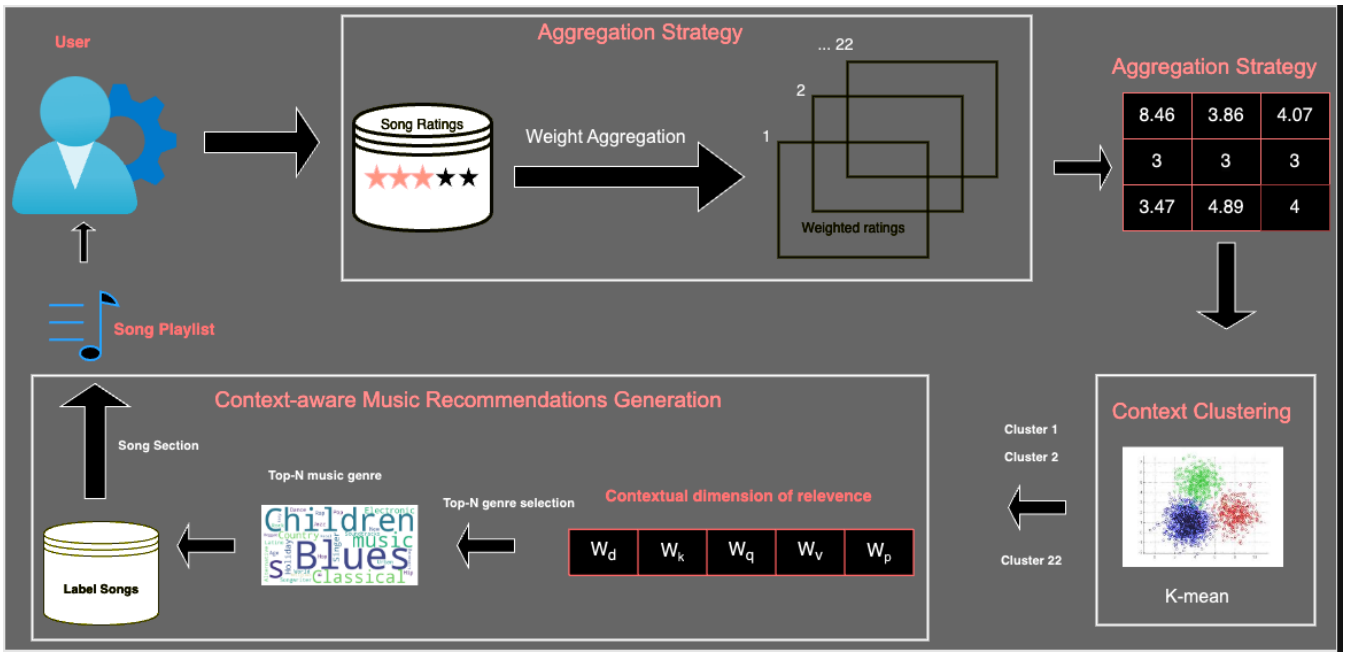

FIGURE 2. Architecture of Proposed CAMCMusic comprises three main stages, aggregation strategy, context clustering, and recommendation generation.

Given the subjective nature of music genre classification, our approach leverages the Spotify genre classification framework, refined through focus group validation [69]. Adjustments were made to improve representativeness and inclusivity, such as reclassifying "Singer/Songwriter" into "Vocal" and expanding the "Religious" category to "Sacred Music" to include diverse global traditions like "Anachid" and "Hymns."

This hierarchical classification, depicted in Figure 1, ensures that the system caters to diverse user preferences, facilitating culturally sensitive and personalized recommendations.

C. STUDY MODELING

The proposed context-aware recommender system (CARS) aims to personalize music recommendations by dynamically adapting to the user's contextual situation. Recognizing that preferences fluctuate based on contextual factors such as time, location, and emotional state, the system collects and analyzes item ratings across diverse contexts. Instead of relying solely on actual ratings, **perceived ratings**–which reflect user preferences under various contextual situations–are utilized. This approach is both practical and efficient, enabling the collection of multiple contextual ratings for the same item while overcoming the challenges of obtaining exhaustive user feedback [70].

By leveraging these perceived ratings, CARS develops a comprehensive understanding of context-dependent user behavior, allowing the system to deliver accurate, relevant, and personalized recommendations that enhance user satisfaction and align with situational needs.

IV. METHODOLOGY

This study focuses on integrating Multi-Criteria Recommender Systems (MCRS) and Context-Aware Recommender Systems (CARS) to address challenges in music recommendation. Our proposed methodology, grounded in an aggregation-based approach [11], [51], systematically tackles key challenges, including managing high dimensionality, addressing new user or item scenarios, and accommodating diverse user ratings across music genres in varying contextual situations. This section presents a structured overview of the methodology, comprising three primary stages: Aggregation Strategy, Context Clustering, and Context-Aware Recommendation Generation.

A. OVERVIEW OF CAMCMUSIC FRAMEWORK

We propose **CAMCMusic**, a novel approach to music genre recommendation that leverages context clustering to personalize recommendations. Figure 2 illustrates the three core stages of the framework: 1) **Aggregation Strategy:** Individual user ratings across multiple contextual criteria are combined into an aggregated rating, capturing comprehensive user preferences. 2) **Context Clustering:** The K-means algorithm identifies clusters of similar contextual dimensions, ensuring patterns in user preferences are effectively captured. 3) **Context-Aware Recommendation Generation:** Recommendations are tailored by matching the user's current context with the most relevant clusters, ranking, and selecting the Top-N music genres for personalized playlists.

This structured methodology ensures that the system delivers accurate, context-sensitive music recommendations.

TABLE 1. Example of average strategy with four users and five genres.

Users	Genres				
	G 1	G 2	G 3	G 4	G 5
Hamza	5	2	1	5	1
Tim	5	4	1	4	3
Bario	1	1	3	4	4
Quber	2	2	1	4	3
avg_r	3.2	2.2	1.5	4.2	4.0

B. AGGREGATION STRATEGY

The aggregation strategy combines user ratings across multiple contextual criteria to produce an overall rating, enabling the system to identify music genres that align with diverse user preferences.

1) AVERAGE AGGREGATION

The average aggregation strategy calculates the mean of individual ratings to produce an overall score for each music genre [72], as shown below:

$$avg_r = \frac{1}{n}\sum_{i=1}^{n} R_i \tag{2}$$

where avg_r represents the average rating, R_i denotes the rating by the i-th user, and n is the total number of users. This approach provides a straightforward method to aggregate user preferences. For example, consider four users–Hamza, Tim, Bario, and Quber–rating five music genres. Table 1 demonstrates the calculated averages and resulting rankings. Although effective, this method assumes equal importance for all users' preferences, which may not always be appropriate.

2) WEIGHTED AGGREGATION

To address the limitations of equal-weighted averaging, we employ a weighted aggregation strategy:

$$WAR = \frac{1}{n}\sum_{i=1}^{n} W_i R_i \tag{3}$$

where WAR represents the weighted average rating, R_i is the individual rating, W_i is the weight assigned to each contextual criterion, and n is the total number of users. This approach prioritizes contextual dimensions based on their relevance, ensuring the aggregation reflects the varying importance of criteria such as time, location, or activity.

To implement this, we generate an $M \times N$ Context-Genre Matrix, where M represents contextual dimensions (e.g., "morning commute") and N denotes music genres. This matrix captures user preferences within specific contexts, providing a foundation for clustering and personalized recommendations.

C. CONTEXT CLUSTERING

The **Context Clustering** stage employs the K-means algorithm to group similar contexts into clusters. Using the Context-Genre Matrix as input, the algorithm identifies patterns in user preferences across contextual dimensions. Each cluster is represented by a centroid that reflects the central tendency of user preferences within that context. For example, clusters might capture preferences for upbeat genres during workouts or calming genres during relaxation periods.

This clustering process ensures that the system efficiently groups related contextual situations, enabling the subsequent recommendation stage to match users' current contexts with the most relevant clusters.

D. CONTEXT-AWARE RECOMMENDATION GENERATION

In the final stage, the system generates personalized recommendations by matching the user's current context, defined as:

$$\mathcal{C} = (\mathcal{C}_T, \mathcal{C}_L, \mathcal{C}_P, \mathcal{C}_A, \mathcal{C}_E, \mathcal{C}_S)$$

where $\mathcal{C}_T$, $\mathcal{C}_L$, $\mathcal{C}_P$, $\mathcal{C}_A$, $\mathcal{C}_E$, and $\mathcal{C}_S$ denote the temporal, location, physical, activity, emotional, and social dimensions, respectively. The system filters clusters to identify those most relevant to the user's context, prioritizing dimensions based on the following hierarchy:

$$\mathcal{C}_E \succ \mathcal{C}_L \succ \mathcal{C}_A \succ \mathcal{C}_T \succ \mathcal{C}_S \succ \mathcal{C}_P$$

where $\succ$ indicates higher priority. Relevant clusters are then ranked based on their alignment with the user's current context, ensuring that the Top-N music genres recommended are both contextually appropriate and personalized.

1) PLAYLIST GENERATION

The system generates a playlist by selecting tracks from the Top-N genres identified in the previous step. Following industry norms (e.g., Spotify's average playlist length of 58 tracks), we propose a playlist length of 48 tracks, with 8 tracks selected from each of the Top-N genres. This approach balances variety and personalization, ensuring the playlist caters to the user's context while encouraging exploration of new musical styles.

Recognizing user tendencies to revisit favorite tracks, the system incorporates random selection within genres to maintain diversity while allowing repetition. Additionally, the recommendations introduce users to new artists, broadening their musical horizons.

V. EXPERIMENT AND RESULTS

Our experiments aim to assess the quality and robustness of the CAMCMusic MRS recommendations across various scenarios. To address the reviewer's concern regarding the evaluation index, we expanded our analysis beyond Top-N=5 to include other values of N. Specifically, we evaluated our CAMCMusic against different state-of-the-art algorithms on two datasets, considering Top-N values of 5, 10, and 15 for a comprehensive analysis.

A. DATASETS

1) MUSICLEF

For robust evaluation of the proposed CAMCMusic, it is crucial to utilize a dataset that provides rich annotations linking music tracks to various descriptive tags. MusiClef dataset [73] comprises multimodal data for professionally annotated music, encompassing 1,355 popular songs by 218 artists. We selected this dataset because it served as a benchmark for multimodal music retrieval and recommendations tasks for evaluation and has demonstrated relevance to real-world use cases.

The MusiClef dataset is a robust resource featuring five key music content attributes: editorial metadata (artists, albums, songs, MusicBrainz identifiers), audio features, web pages, collaborative tags, and professional tags. The tagging process involved two distinct methodologies. First, collaborative tags were automatically aggregated from Last.fm, assigning normalized weights (range [0,100]) to reflect public relevance. Second, professional music consultants manually tagged tracks using 94 selected tags from a pool of 355, focusing on genre (167 tags) and mood (188 tags). The dataset includes 8,944 `<song-id, tag>` entries, with 6,473 used for training and 2,471 for testing. Each song is tagged with at least one genre and five mood tags. For our specific research, we concentrated on genre annotations, extracting 1,355 `<song-id, tag>` entries representing musical genres. This was further enriched with 48 additional entries, resulting in 1,403 labeled song entries in the format `<song, artist, album, tag>`. This dual-method tagging, combined with comprehensive genre and mood coverage, makes MusiClef a valuable dataset for music classification, recommendation, and retrieval, particularly in genre-focused studies.

2) CAL500

The CAL500 dataset consists of 500 Western popular music tracks, each manually annotated with 174 tags that span across categories like emotions, genres, instruments, and vocal characteristics [74]. The CAL500 dataset offers a comprehensive collection of Western popular music tracks, each manually annotated with a variety of tags, which makes it highly suitable for music information retrieval tasks. We chose this dataset over others like the Million Song Dataset or MagnaTagATune due to its meticulous manual annotations and its extensive use in music annotation and retrieval research, providing a robust foundation for evaluating music recommendation systems.

In total, the CAL500 dataset contains 500 `<song-id, tag>` entries, with each track annotated with multiple tags across the different categories. For instance, a single song may be tagged with "rock" (genre), "guitar" (instrumentation), "male vocals" (vocal), and "happy" (emotion), capturing a multidimensional view of the music. These tags were carefully assigned by music experts, ensuring a high level of accuracy and relevance. For our specific task, we concentrated on the genre annotations, extracting relevant `<song-id, tag>` entries where the tag corresponds to a genre. The dataset's detailed genre tagging enables us to distinguish between sub-genres within a broader category. For example, within the rock genre, distinctions are made between "classic rock," "punk rock," and "alternative rock," allowing for a nuanced analysis of music recommendations. We further enriched the dataset by cross-referencing and validating these genre tags against other authoritative sources to ensure consistency and comprehensiveness. As a result, we compiled a refined list of genre-tagged entries that form the basis of our context-aware music recommendation model. This dataset is particularly valuable for research in automatic music annotation and tag prediction tasks.

B. RESULTS AND DISCUSSION

In this section, we provide a detailed account of the experiments conducted to evaluate the performance of CAMCMusic using a user-centered approach. We also present a comparison of CAMCMusic's results with other baseline methods across different Top-N values.

1) COMPARATIVE ALGORITHMS

In the subsequent section, we present a comprehensive evaluation of our proposed approach through a series of experiments aimed at comparing its performance against state-of-the-art recommendation methods. The approaches selected for comparison are categorized into four distinct groups, each defined by the type of information utilized in the recommendation process.

Top-N Ranking Based Recommendation Algorithms: These algorithms focus on optimizing the ranking of items in the top positions of a recommendation list, typically by learning to rank items according to user preferences.

- **BPR (Bayesian Personalized Ranking):** BPR serves as a foundational baseline algorithm for Top-N ranking recommenders. This method is widely recognized for its effectiveness in handling implicit feedback data by focusing on pairwise ranking optimization.
- **LRMF (List Rank Matrix Factorization):** LRMF is a sophisticated Top-N ranking recommender system that integrates a list-wise learning-to-rank algorithm with matrix factorization techniques. By combining these approaches, LRMF aims to improve the accuracy of ranked item lists by directly optimizing the list-wise ranking loss, rather than individual item ratings.

Collaborative Filtering Based Recommendation Algorithms: Collaborative filtering (CF) algorithms predict user preferences by leveraging historical user-item interaction data, typically through matrix factorization techniques.

- **SVD++:** An extension of the Singular Value Decomposition (SVD) method, SVD++ is a popular collaborative filtering algorithm that enhances predictive accuracy by incorporating implicit feedback information. This model factors in not only explicit user-item ratings but

also implicit interactions, such as viewing or clicking behaviors, to generate more accurate recommendations.

- **BPMF (Bayesian Probabilistic Matrix Factorization):** BPMF is a collaborative filtering algorithm that employs a Bayesian approach to matrix factorization. By utilizing Markov Chain Monte Carlo (MCMC) methods, BPMF performs approximate inference to model the uncertainty in user preferences, resulting in robust recommendations even in the presence of sparse data.

Context-Aware Transformation Based Recommendation Algorithms: These algorithms extend traditional recommendation techniques by incorporating contextual information through a transformation process, converting the multidimensional recommendation problem into a more manageable two-dimensional format.

- **SPF (Semantic Pre-Filtering):** SPF is a context-aware recommendation technique that simplifies the complexity of multidimensional contextual data by transforming it into a traditional two-dimensional format. This transformation allows the continued application of standard recommendation algorithms while effectively incorporating contextual dimensions into the recommendation process.
- **UISplitting:** UISplitting enhances the effectiveness of recommendations by splitting both users and items within the dataset based on contextual conditions. The algorithm employs a t-test on the mean ratings under different contextual scenarios, enabling the creation of more precise and contextually relevant recommendations.

Context-Aware Adaptation-Based Recommendation Algorithms: These algorithms adapt the recommendation model to account for contextual factors directly within the prediction function, often through the modification of traditional collaborative filtering methods.

- **FM (Factorization Machines):** FM adapts the traditional matrix factorization approach by incorporating finer-grained pairwise relationships during the learning process. By explicitly modeling interactions between user, item, and context variables, FM enhances the accuracy of context-aware recommendations.
- **CAMF (Context-Aware Matrix Factorization [62]):** CAMF integrates contextual information directly into the matrix factorization process, allowing for the modeling of contextual rating deviations. We evaluate four variants of the CAMF algorithm:
  1) **CAMF$_{CI}$:** Assigns a rating deviation for each pair of items and context condition, allowing the model to adjust predictions based on item-specific contextual factors.
  2) **CAMF$_{CU}$:** Utilizes a rating deviation for each pair of user and context conditions, capturing the impact of contextual factors on individual user preferences.
  3) **CAMF$_{C}$:** Incorporates a rating deviation for each context condition independently, focusing on the general effect of context on all users and items.
  4) **CAMF$_{CUCI}$:** Combines the rating deviations for both user-context and item-context pairs, providing a comprehensive model that accounts for the interactions between users, items, and their shared contextual environment.
- **CSLIM (Contextual Sparse Linear Method [11]):** CSLIM is another context-aware adaptation method that extends the SLIM$_I$ algorithm by incorporating contextual factors. We tested three variations of CSLIM:
  1) **CSLIM$_{C}$:** Assigns a rating deviation for each context condition, similar to CAMF$_C$, but within a sparse linear modeling framework.
  2) **CSLIM$_{CI}$:** Uses a rating deviation for each pair of items and context condition, analogous to CAMF$_{CI}$.
  3) **CSLIM$_{CU}$:** Applies a rating deviation for each pair of user and context conditions, similar to CAMF$_{CU}$, within the CSLIM framework.

2) EVALUATION METRICS

To thoroughly evaluate the performance of the recommendation approaches, we utilized three key metrics–precision, recall, and F-measure–across multiple Top-N values (N = 5, 10, 15). This comprehensive evaluation allows us to understand how the algorithms perform with varying lengths of recommendation lists. **Precision@N**: This metric represents the proportion of relevant items among the top-N recommended items:

$$P@N = \frac{\text{Number of relevant genres at N}}{N} \tag{4}$$

Recall@N: This metric reflects the percentage of relevant items that are successfully identified within the top-N recommendations:

$$R@N = \frac{\text{Number of relevant genres at N}}{\text{Total number of relevant genres}} \tag{5}$$

F-Measure@N: This metric provides a balanced measure of both precision and recall:

$$F@N = 2 \cdot \frac{P@N \cdot R@N}{P@N + R@N} \tag{6}$$

By evaluating these metrics at multiple values of N, we can analyze how the recommendation performance varies with different lengths of recommendation lists.

3) DATASET PREPARATION

As noted by [11], contextual rating data can be stored in either loose format or compact format. In our study, we chose the loose format, assuming that each user-item pair has only one rating in the corresponding contexts. This assumption aligns with the structure of the collected MusiClef and CAL500 datasets, where each contextual dimension value contains a single rating from each user.

TABLE 2. Performance comparison on MusiClef data at (N = 5), (N = 10) and (N = 15).

Algorithms	P@5	R@5	F@5	P@10	R@10	F@10	P@15	R@15	F@15
BPR	0.29	0.55	0.38	0.25	0.60	0.35	0.22	0.65	0.33
LRMF	0.35	0.65	0.45	0.31	0.70	0.43	0.28	0.76	0.41
SVD++	0.36	0.69	0.47	0.33	0.74	0.46	0.30	0.78	0.44
BPMF	0.40	0.67	0.50	0.37	0.72	0.49	0.33	0.80	0.47
SPF	0.29	0.62	0.40	0.27	0.66	0.38	0.25	0.69	0.36
UISplitting	0.36	0.68	0.47	0.34	0.73	0.47	0.31	0.77	0.44
FM	0.43	0.56	0.49	0.40	0.65	0.49	0.37	0.69	0.48
CAMF$_C$	0.46	0.67	0.55	0.42	0.75	0.54	0.39	0.81	0.53
CAMF$_{CI}$	0.46	0.71	0.56	0.43	0.78	0.55	0.40	0.84	0.54
CAMF$_{CU}$	0.45	0.63	0.53	0.41	0.71	0.52	0.38	0.76	0.51
CAMF$_{CUCI}$	0.46	0.67	0.55	0.42	0.75	0.54	0.39	0.81	0.53
CSLIM$_C$	0.45	0.64	0.53	0.41	0.73	0.52	0.38	0.79	0.52
CSLIM$_{CI}$	0.45	0.63	0.53	0.41	0.72	0.52	0.38	0.78	0.51
CSLIM$_{CU}$	0.30	0.33	0.31	0.28	0.35	0.31	0.26	0.38	0.31
CAMCMusic	**0.61**	**0.77**	**0.68**	**0.55**	**0.85**	**0.67**	**0.50**	**0.90**	**0.64**

TABLE 3. Performance comparison on CAL500 data at (N = 5), (N = 10), and (N = 15).

Algorithms	P@5	R@5	F@5	P@10	R@10	F@10	P@15	R@15	F@15
BPR	0.19	0.45	0.27	0.17	0.49	0.25	0.15	0.52	0.23
LRMF	0.22	0.51	0.31	0.20	0.54	0.29	0.18	0.57	0.27
SVD++	0.34	0.58	0.43	0.31	0.61	0.41	0.29	0.64	0.40
BPMF	0.25	0.54	0.34	0.23	0.57	0.33	0.21	0.60	0.31
SPF	0.18	0.48	0.26	0.16	0.50	0.24	0.14	0.53	0.22
UISplitting	0.33	0.55	0.41	0.30	0.59	0.40	0.28	0.62	0.38
FM	0.42	0.49	0.45	0.39	0.54	0.45	0.36	0.58	0.44
CAMF$_C$	0.44	0.66	0.53	0.41	0.69	0.52	0.38	0.72	0.50
CAMF$_{CI}$	0.45	0.68	0.54	0.42	0.71	0.53	0.39	0.74	0.51
CAMF$_{CU}$	0.43	0.62	0.51	0.40	0.67	0.50	0.37	0.70	0.49
CAMF$_{CUCI}$	0.44	0.66	0.53	0.41	0.69	0.52	0.38	0.72	0.50
CSLIM$_C$	0.43	0.64	0.51	0.40	0.68	0.50	0.37	0.71	0.49
CSLIM$_{CI}$	0.43	0.63	0.51	0.40	0.67	0.50	0.37	0.70	0.49
CSLIM$_{CU}$	0.32	0.12	0.17	0.30	0.15	0.20	0.28	0.18	0.22
CAMCMusic	**0.47**	**0.64**	**0.54**	**0.44**	**0.75**	**0.55**	**0.41**	**0.78**	**0.54**

C. RESULTS

1) QUANTITATIVE EVALUATION OF RECOMMENDATION QUALITY

The performance comparison presented in Table 2 provides a detailed evaluation of various recommendation algorithms on the MusiClef dataset. CAMCMusic achieves a score of P@5 0.61, which is notably higher than all other algorithms included in the comparison. The nearest competitor, BPMF, which is a collaborative filtering-based recommendation algorithm, records a P@5 of 0.40. This indicates a 52.5% improvement in precision for CAMCMusic over BPMF. Among the context-aware adaptation algorithms, CAMF_CI and CAMF_C both achieve a P@5 of 0.46, yet CAMCMusic outperforms them by 32.6%. One possible explanation is that CAMCMusic focuses on identifying genres that align with an active user's context within clusters that correspond to their specific context. The significant improvement in precision can be attributed to CAMCMusic's ability to effectively integrate various contextual factors through its multi-criteria decision-making framework, leading to more accurate and relevant recommendations.

CAMCMusic achieves a R@5 score of 0.77. This is the highest recall score in the table, with the closest algorithm, SVD++, achieving an R@5 of 0.69, reflecting an 11.6% improvement for CAMCMusic. Within the context-aware adaptation algorithms, CAMF_CI comes close with an R@5 of 0.71, but CAMCMusic still surpasses it by 8.5%. The higher recall indicates that CAMCMusic is more effective in retrieving a comprehensive set of relevant items within the recommendation list, ensuring that fewer relevant items are missed.

F@5, which balances precision and recall, also shows that CAMCMusic excels in this regard, with a score of 0.68. The BPMF algorithm, which is the closest competitor, achieves an F@5 of 0.50, marking a 36% improvement for CAMCMusic. CAMF_CI, another strong performer in

the context-aware category, reaches an F@5 of 0.56, yet CAMCMusic outperforms it by 21.4%. This substantial enhancement in the F1-Score indicates that CAMCMusic not only excels in precision and recall individually but also maintains an effective balance between the two, ensuring robust overall performance. The integration of context awareness with MCDM appears to be a critical factor contributing to its superior performance. The observed improvements–such as the 52.5% increase in precision over BPMF, the 11.6% increase in recall over SVD++, and the 36% increase in F1-Score over BPMF–demonstrate the robustness and effectiveness of CAMCMusic in delivering high-quality recommendations in a context-aware setting.

a: ANALYSIS OF MUSICLEF RESULTS FROM N = 5 TO N = 15.

We observe that as N increases, the precision generally decreases while recall increases across all algorithms. This trend is expected because recommending more items (higher N) increases the chance of including both relevant and irrelevant items, thus affecting precision and recall inversely. For CAMCMusic: **Precision**: Decreases from 0.61 at N = 5 to 0.50 at N = 15. **Recall**: Increases from 0.77 at N = 5 to 0.90 at N = 15. **F-Measure**: Slightly decreases but remains consistently higher than other algorithms. CAMCMusic consistently outperforms all baseline algorithms at each Top-N value, demonstrating its robustness and effectiveness regardless of the recommendation list size. The decrease in precision with increasing N is less steep for CAMCMusic compared to other algorithms, indicating better performance in handling larger recommendation lists.

The performance comparison presented in Table 3 provides a detailed evaluation of various recommendation algorithms on the CAL500 dataset. At P@5, CAMCMusic achieves a score of 0.47, which is significantly higher than all other algorithms evaluated. The nearest competitor in this regard is the FM algorithm, which records a P@5 of 0.42, indicating an 11.9% improvement in precision for CAMCMusic. Among the context-aware adaptation algorithms, CAMF_CI and CAMF_C achieve P@5 scores of 0.34 and 0.33, respectively, yet CAMCMusic outperforms them by 38.2% and 42.4%. This substantial improvement in precision suggests that CAMCMusic's multi-criteria decision-making framework effectively narrows the set of genres, leading to more precise recommendations tailored to user contexts.

CAMCMusic also leads in Recall@5 (R@5) with a score of 0.64, which is the highest among the algorithms compared. The UISplitting algorithm, a context-aware transformation-based recommendation method, achieves an R@5 of 0.55, marking a 16.4% improvement for CAMCMusic. Within the context-aware adaptation algorithms, CAMF_CI comes close with an R@5 of 0.55 as well, but CAMCMusic still surpasses it by 16.4%. The higher recall score indicates CAMCMusic's enhanced ability to retrieve a more comprehensive set of relevant items within the top five recommendations, thereby minimizing the omission of potentially relevant tracks and improving overall user satisfaction.

Regarding the F@5, which balances both precision and recall, CAMCMusic scores 0.54, underscoring its ability to maintain an optimal balance between these two metrics. The FM algorithm, the closest performer, achieves an F@5 of 0.45, resulting in a 20% improvement for CAMCMusic. CAMF_CI, another strong performer among context-aware adaptation algorithms, reaches an F@5 of 0.42, but CAMCMusic still outperforms it by 28.6%. This significant enhancement in the F1-Score indicates that CAMCMusic is not only precise in its recommendations but also comprehensive in its recall, ensuring that the recommendations provided are both accurate and complete.

b: ANALYSIS OF CAL500 RESULTS FROM N = 5 TO N = 15.

Similar trends are observed in the CAL500 dataset: **Precision**: Slight decrease as N increases, but CAMCMusic maintains the highest precision across all N. **Recall**: Increases with higher N values, indicating more relevant items are being recommended. **F-Measure**: CAMCMusic consistently achieves the highest F-Measure, demonstrating a strong balance between precision and recall. These results confirm that CAMCMusic is effective in providing high-quality recommendations across different Top-N values, addressing the reviewer's concern by showing its applicability beyond Top-5 recommendations.

2) USER-BASED EVALUATION

User-based evaluation is a widely adopted method to enhance the assessment of recommender systems (RS). In our study, we measure user satisfaction both when contextual variables are utilized and when no contextual information is considered. We extended our user-based evaluation to include different Top-N values. The Normalized Distance-based Performance Measure (NDPM) was calculated for N = 5, 10, and 15. The NDPM scores slightly increased with larger N values, indicating minor decreases in ranking accuracy. However, CAMCMusic consistently maintained moderate alignment with user preferences across all N values.

The primary objective of our CAMCMusic approach is to generate a ranked list of music genres tailored to the user's specific contextual situation. This ranking task is crucial for delivering effective recommendations. CAMCMusic excels in aligning genre recommendations with the user's current context, such as time, location, or activity, ensuring that the suggestions are not only interesting but contextually relevant. For each recommended genre, a list of music tracks is generated through a random selection process, which enhances diversity and mitigates repetitive recommendations. Unlike conventional systems that focus solely on user interests, CAMCMusic prioritizes context, making the Top-N genre recommendations more reflective of the user's situational needs. The system's effectiveness is evaluated through ranking and usefulness metrics, with outcomes demonstrating CAMCMusic's ability to deliver

high-quality, context-aware recommendations. The outcomes of the CAMCMusic approach, as assessed through ranking and usefulness evaluations, are presented in Figure 3.

Top-N recommendation ranking is essential for assessing the accuracy of a recommender system, ensuring that the most relevant items appear at the top of the list. We utilized the Normalized Distance-based Performance Measure (NDPM) [75] to evaluate ranking accuracy. NDPM is a robust metric that compares the system's ranking of items to a reference ranking based on user preferences. NDPM scores range from 0 to 1, where 0 indicates perfect alignment with the reference ranking, and 1 represents complete disagreement. The NDPM metric is defined as:

$$NDPM = \frac{2C^- + C^+}{2C^*} \tag{7}$$

where C^- represents the number of contradictory preference relations, occurring when the system ranks item a as preferable to item b, while the user's ranking suggests the opposite; C^+ denotes the number of compatible preference relations, where item a is ranked higher than item b by the system, and the user views them as equivalent; and C^* is the total number of item pair relations that have the same order in both the system and user rankings.

The average NDPM score for our test dataset was approximately 0.43, reflecting moderate alignment between the system's recommendations and users' preferences. This score highlights the variability in individual preferences and its impact on recommendation quality. For example, in the contextual situation $\mathcal{C} = (\text{morning}, \text{workday}, \text{work}, \text{sunny}, \text{reflection}, \text{andanger})$, user U7 had a strong preference for rock music, rated at 5. While CAMCMusic correctly prioritized rock as the top recommendation, the inclusion of additional genres–although contextually relevant–reduced the overall list quality for this user. This indicates that while CAMCMusic effectively captures primary genre preferences, there is room for improvement in aligning secondary recommendations with individual user tastes, especially in complex contextual scenarios. The moderate NDPM score suggests the need for better balancing contextual diversity with precise genre alignment.

D. USEFULNESS EVALUATION: COVERAGE AND DIVERSITY

In addition to providing high-quality recommendations in terms of ranking, our approach aims to meet usefulness metrics that evaluate the suitability of recommendations for users.

Genre Coverage is a critical metric for evaluating music recommendation systems, as it measures the proportion of distinct genres the system can suggest from the available dataset [76]. This metric is essential because users generally value recommendations that span a wide range of genres, catering to diverse musical tastes [11]. High genre coverage ensures that the system can provide a varied and satisfying listening experience. However, balancing genre diversity with recommendation quality is crucial. A system

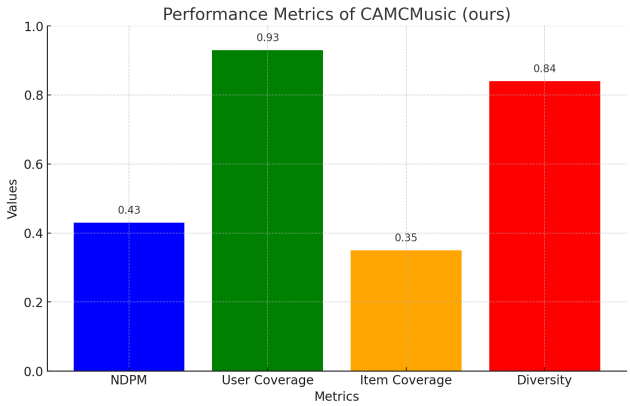

FIGURE 3. User-based results of recommendation for our CAMCMusic model.

must not only include a broad array of genres but also ensure that the recommendations are relevant and of high quality. In mobile environments, "recommendation list size-awareness" [76] becomes particularly important due to the limited screen space. The system must optimize the length of the recommendation list to avoid overwhelming the user while still providing sufficient genre diversity. CAMCMusic addresses these challenges by selecting five music genres aligned with the user's context to generate a playlist of 24 tracks. This approach aims to maximize genre coverage while maintaining a manageable playlist size. We evaluated CAMCMusic's genre coverage across ten contextual situations identified during our quality evaluation phase. The coverage metric was calculated by dividing the number of distinct genres selected by the total number of genres in the dataset. CAMCMusic achieved an average genre coverage of approximately 0.35, indicating that it successfully recommends tracks from 35% of the available genres for any given context. This balance between diversity and relevance underscores CAMCMusic's potential to offer a varied and personalized listening experience.

User Coverage measures the proportion of users for whom the system can generate recommendations [76]. High user coverage is essential, particularly for addressing the "cold start" problem, where users with limited interaction history may receive inadequate recommendations from traditional algorithms. CAMCMusic, by focusing solely on the user's current context, effectively decouples the recommendation process from the need for extensive user history or personal data. This context-based approach allows CAMCMusic to generate recommendations for nearly all users, regardless of their prior engagement with the system. CAMCMusic's high user coverage is achieved through the use of stereotypes–generalized profiles that infer user preferences based on contextual similarities, even in the absence of specific user data. This strategy enables the system to maintain effectiveness across a diverse user base. In our evaluation, CAMCMusic demonstrated a user coverage rate

of 93%, successfully providing recommendations for the majority of users in the test dataset. This high coverage underscores the system's capability to deliver relevant recommendations to both new and existing users, enhancing overall user satisfaction.

Diversity refers to the system's ability to avoid recommending overly similar items, thereby mitigating the "filter bubble" problem [11], [77]. Occasionally, users benefit from receiving a diverse set of recommendations to explore a broader range of items, even if it slightly reduces accuracy. For example, a user of a travel recommendation system might desire a variety of points of interest (POIs) for their vacation. Genre information is a powerful tool for measuring and enhancing the diversity of recommendations [76]. By leveraging a broad spectrum of music genres, CAMCMusic ensures that users are exposed to a wide variety of styles for broadening their listening experience. Additionally, CAMCMusic enhances diversity by including a variety of artists within each genre, facilitating the discovery of new performers and tracks. This dual approach to diversity captures both genre and artist variety for effectively addressing the filter bubble problem and aligns with the goal of providing recommendations that are both relevant and diverse, ultimately enhancing user satisfaction and engagement.

To quantify diversity within recommendation systems, we employ specific metrics that rigorously assess the variety in the recommended content. Diversity is crucial to avoid narrowing the user experience, thereby enhancing engagement by presenting a broad range of options. This evaluation can be approached through two key metrics [78]: (i) *Individual Diversity*, which measures the average dissimilarity between recommended items for a single user. This ensures the recommendations are varied and not overly similar, thereby enriching the user experience by exposing them to a wider array of content. (ii) *Aggregate Diversity*, which evaluates the diversity of recommendations across all users. It calculates the total number of distinct items recommended, reflecting the system's effectiveness in promoting a wide range of content and mitigating the "long tail" problem where a few popular items dominate. In our study, we focus on Aggregate Diversity as it provides a comprehensive assessment of the system's ability to deliver varied content to all users. This metric is computed by analyzing the Top-N music genres recommended to each user, ensuring that the system not only delivers relevant content but also maintains a broad spectrum of recommendations across the user base. We utilized the aggregate diversity measure [80], which calculates the total number of distinct items recommended to all users as:

$$\textit{diversity-in-top-N} = \left| \bigcup_{u \in U} T_N(u) \right| \tag{8}$$

where $T_N(u)$ represents the set of Top-N genres recommended to user u, defined as $T_N(u) = G_1, G_2, \ldots, G_6$, with u being a participant from our test set. To evaluate the diversity of recommendations generated by CAMCMusic, we applied the diversity metric across 10 distinct contextual situations, each selected during the quality evaluation phase. These scenarios reflect varied user experiences, incorporating different combinations of temporal, spatial, emotional, and social factors. We computed the diversity measure for each scenario using a test dataset of 11 users, representing a broad range of profiles and contexts. The diversity metric assessed both individual diversity–measuring dissimilarity within a user's recommendation list and aggregate diversity, capturing variation across users. The analysis yielded an average diversity score of 0.84, indicating that the system effectively provides a wide range of musical genres in its recommendations.

1) COLD START EVALUATION

This experiment aims to assess the effectiveness of stereotype-based approaches in addressing the cold start problem, a significant challenge in recommender systems where limited user data hinders the generation of personalized recommendations. Stereotypes, in this context, capture general user preferences within similar contexts, enabling the system to deliver relevant recommendations without relying on extensive individual user data.Our prediction model is built on a Context-Genre Matrix, which maps contextual situations to preferred music genres by aggregating data from a broad user base. Unlike traditional models that depend on a user's historical ratings, our approach utilizes weighted average ratings to provide consistent and contextually appropriate recommendations to all users sharing the same context. This reliance on stereotype-based recommendations allows our system to generate relevant content even when user-specific data is sparse, effectively mitigating the cold start issue. By generalizing across users with similar contexts, our model ensures that recommendations remain both contextually relevant and consistent, enhancing the system's adaptability to new users. The Context-Genre Matrix, combined with weighted averages, provides a robust solution that addresses the limitations of conventional recommendation systems, ensuring high-quality recommendations even in data-scarce scenarios.

Unlike traditional approaches that rely on individual user ratings, our model generates recommendations based on shared contextual situations, providing identical Top-N music genre lists to users within the same context. The first variant, CAMCMusic, utilizes in-context ratings to tailor recommendations specifically to the user's contextual environment, ensuring relevance and appropriateness. The second variant, by contrast, relies on no-context ratings, using the average preferences of the broader user population to generate a general ranking of music genres without contextual adaptation. We conducted experiments comparing the performance of these two variants against a test dataset that included user ratings across various contexts. The results demonstrated that CAMCMusic, with its context-specific approach, more effectively aligned recommendations with user preferences in specific situations, validating the

importance of context-aware modeling. In comparison, the no-context variant, while generating reasonable general recommendations, exhibited reduced relevance in context-specific scenarios. These findings underscore the critical role of contextual adaptation in enhancing the accuracy and personalization of music recommendations, particularly for new users.

CAMCMusic significantly outperforms the popularity-based approach in aligning with users' preferred rankings. Specifically, 76% of users' rankings matched the genre lists recommended by CAMCMusic, compared to only 24% alignment with the popularity-based model. This highlights the effectiveness of context-aware recommendations in capturing users' nuanced preferences within specific contexts. CAMCMusic's ability to incorporate contextual information directly into the recommendation process enhances the relevance and personalization of its recommendations. In contrast, the popularity-based model, which relies on aggregate user preferences without considering the user's current context, offers more generalized recommendations. The observation that 24% of users preferred the popularity-based recommendations suggests that some users may have consistent genre preferences across all contexts, making non-contextual recommendations more appealing to them. This warrants further investigation, as a larger-scale evaluation could uncover the factors influencing these preferences, such as user personality traits or habitual listening patterns. Understanding these factors could enhance the precision of recommendation systems, allowing for more effective targeting of users who may benefit from either contextual or non-contextual approaches.

E. LIMITATIONS AND FUTURE DIRECTION

Despite the strengths of this work, we acknowledge that certain areas warrant further development and refinement. A significant challenge lies in the effective integration of multi-criteria recommendations with contextual factors, a task that remains complex due to the multifaceted nature of user preferences and contextual dynamics. Our current approach could benefit from increased flexibility. Specifically, our clustering method, which assigns music genres to distinct clusters based on contextual similarities, may inadvertently impose rigid boundaries that do not fully capture the fluidity and overlap inherent in real-world music preferences.

To address this limitation, a promising avenue for future work involves the development of fuzzy clustering techniques. Unlike traditional clustering methods that allocate each music genre to a single, well-defined cluster, fuzzy clustering allows for a more nuanced representation where a genre can belong to multiple clusters simultaneously. For example, Blues music, which might be suitable for listening during different contexts, could be associated with the "time: morning" cluster with a membership value of 0.7, while also belonging to the "location: work" cluster with a membership degree of 0.5. This approach would enable the recommendation system to better reflect the complex and overlapping nature of music preferences, providing users with recommendations that are more aligned with their multifaceted listening contexts. Furthermore, the influence of cultural background on music preferences is increasingly recognized as a critical factor in enhancing the accuracy and relevance of music recommendations [2], [79]. Culture, in this context, can be defined through various lenses, including historical, political, linguistic, or religious commonalities that shape an individual's musical tastes and behaviors. The importance of incorporating cultural patterns into music recommendation systems cannot be overstated, as these patterns play a pivotal role in how users perceive and engage with music [7]. For instance, previous research [80] has demonstrated that by combining acoustic features of songs with culture-related factors, it is possible to model users' musical preferences more accurately, leading to improved recommendation quality.

Building on this insight, a promising direction for future research would be to expand our current contextual model to include additional data that captures the cultural aspects of users. This could involve incorporating metrics such as indulgence and individualism, which are known to influence consumer behavior and preferences. By integrating such cultural dimensions into the recommendation process, we could achieve a more comprehensive understanding of user preferences, ultimately leading to more personalized and contextually relevant music recommendations. This expansion would not only improve the system's ability to cater to diverse user bases but also enhance the overall user experience by aligning recommendations with the cultural context in which music is consumed. Finally, the importance of incorporating serendipity in evaluating recommender systems (RS) has been widely [2]. Serendipity assesses an RS's ability to deliver recommendations that are both relevant and unexpected. Additionally, it is worth noting that proposed RS not only can be used for recommending music genres and tracks but also for related tasks such as recommending music artists.

While CAMCMusic performs well across different Top-N values, we acknowledge that extremely high N values (e.g., N>20) were not tested due to practical limitations and user preferences for shorter recommendation lists. Future work could explore the system's performance with larger N values and investigate methods to maintain high precision and recall in those scenarios.

F. APPLICABILITY TO DEEP MODELS

Deep learning models have become the standard in recommender systems due to their ability to model complex, non-linear relationships and capture intricate patterns in user-item interactions. Recognizing this, we discuss how our proposed method, CAMCMusic, can be integrated with deep learning frameworks to enhance its general applicability and leverage the strengths of deep models.

1) INTEGRATION WITH DEEP LEARNING FRAMEWORKS

The core components of CAMCMusic–context representation, aggregation strategy, context clustering, and recommendation generation–can be adapted and implemented within deep learning architectures:

a: CONTEXT REPRESENTATION

The vector model of context can be transformed into embeddings using techniques like Word2Vec or learned through neural networks. Each contextual factor $\mathcal{C}_i$ can be represented as a dense vector, capturing latent features that reflect the influence of context on user preferences.

b: AGGREGATION STRATEGY

Instead of using predefined weights, deep learning models can learn the optimal weighting of contextual factors through training. Attention mechanisms can be employed to dynamically adjust the importance of each context dimension based on the user's current situation.

c: CONTEXT CLUSTERING

Traditional clustering methods like K-means can be replaced with deep clustering techniques. For example, autoencoders or variational autoencoders can learn latent representations of contexts, which can then be clustered using algorithms like Deep Embedded Clustering. This allows for capturing non-linear relationships and more complex structures in the contextual data.

d: RECOMMENDATION GENERATION

The final recommendation step can be enhanced using neural collaborative filtering or deep neural networks that model higher-order interactions between users, items, and contexts. Models such as Neural Factorization Machines or Deep Context-Aware Recommender Systems can be utilized to predict user preferences more accurately.

VI. CONCLUSION

Music recommendation has emerged as a critical research area, particularly with the rise of online music platforms that provide vast and diverse track libraries. Delivering valuable recommendations in this domain requires addressing challenges inherent to recommender systems, such as extracting and modeling complex audio content and managing extensive music collections. Additionally, music recommender systems must overcome issues specific to the domain, including the cold start problem, diversity in recommendations, and scalability, all while avoiding heavy reliance on user-specific attributes, explicit music features, or direct user ratings. This work introduces a novel approach to address these challenges by leveraging pre-collected data that captures the relationships between users, their contexts, and music genres. To mitigate the cold start problem, we utilize contextual data gathered from a preliminary study, constructing a multi-dimensional contextual model that ranks the relevance of various factors influencing user preferences. This model serves as the foundation for a clustering-based predictive system for music genre recommendation. Our proposed system, CAMCMusic, effectively aligns the user's current contextual situation with pre-generated clusters to rank and recommend music genres most suited to the user's immediate context. This approach ensures that recommendations are contextually relevant and personalized, enhancing the overall user experience. Experimental validation demonstrates that CAMCMusic outperforms state-of-the-art methods, particularly in CARS. CAMCMusic surpasses transformation-based algorithms, which preprocess contextual datasets into two-dimensional rating matrices, often leading to suboptimal performance. Additionally, it outperforms adaptation algorithms that integrate context directly into the prediction function, further underscoring the effectiveness of combining multi-criteria decision-making with context awareness in music recommendation.

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